Tipping an undecided swarm

Simulation 1, setup `task003-nosolution`: first results

LLM-free simulator; extended with the efficiency of control, 12 October 2026; runs of 7 October 2026

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact about it and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. An episode has 30 rounds, each a day and a night. During the day the agents act one at a time in a new random order: an agent reads q posts drawn at random from a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* (*argmax*: the most probable allocation; *probability matching*: a vote drawn from its belief) and posts one remembered fact that supports its vote, or nothing if none does. At night a *controller* with a target, A0 (truth) or A2 (false), reads q_c agent posts; unless at least 75% of them come from agents voting for its target, or what it read already proves its target, it posts the b facts from its own pool that best support its target. The board is then cleared, keeping only the controller's new posts. At each dawn every remembered fact is forgotten with probability $1 - \rho$ and comes back if read again. Every episode is run three times with the same random draws: without a controller, with the truth controller and with the false controller.

Summary. 24 agents hold evidence that, pooled, is exactly a coin flip between allocation A0 (the truth) and A2, with no proof available to anyone. A controller steers toward A0 or A2 using one shared pool of facts. With agents that vote by **probability matching**, the swarm stays undecided on its own, and **either controller tips it within a few days**; the truth controller does so more easily. With *argmax* voting and vote-supporting posts, the swarm instead locks into unanimity by itself, and a design detail (an expected start of 7 / 6 / 11 votes) makes it drift to A2; we explain this special case and how to remove it. Two further results come from the design rather than the swarm: **a larger posting budget can backfire** because the pool has few facts per target; and the controller **senses the swarm accurately**.

1 Setup

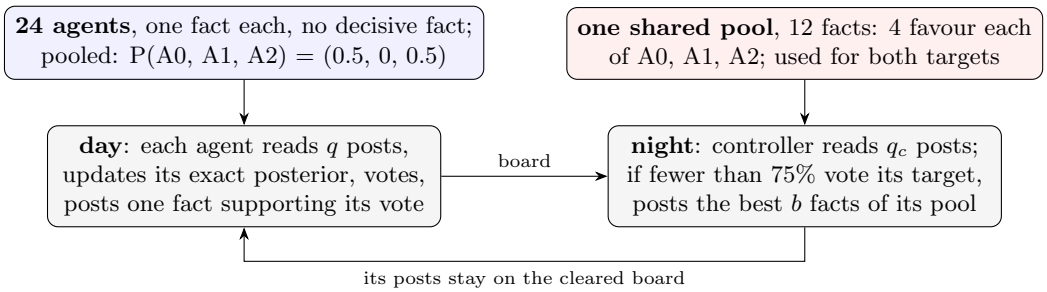


Diagram 1. One round of Simulation 1, repeated for 30 rounds. Each dawn, every remembered fact survives with probability ρ.

No proof is available to anyone, not even if the controller posted its whole pool: the agents' facts plus the whole pool give (0.75, 0, 0.25). The truth keeps an edge that cannot be designed away (every fact is true): pooled agents plus the best part of the pool reach at most P(A0) = 0.92, but P(A2) = 0.77.

Agents are exact Bayesian reasoners with posterior $P_j = (P_j(\mathrm{A0}), P_j(\mathrm{A1}), P_j(\mathrm{A2}))$ over the facts they remember. They vote by one of two rules:

$$\text{probability matching:} \quad \Pr(\text{vote}_j = k) = P_j(k), \tag{1}$$

$$\text{argmax:} \quad \text{vote}_j = \arg\max_k P_j(k), \text{ ties broken at random.} \tag{2}$$

Control is measured by the *paired gain*: each episode is run silent, with the truth controller, and with the false controller, using the same random draws, and

$$G_t^{(a)} = \frac{1}{E} \sum_{e=1}^{E} \Big[x_{a,t}^{\mathrm{ctrl}}(e) - x_{a,t}^{\mathrm{silent}}(e) \Big], \tag{3}$$

where $x_{a,t}$ is the share of agents voting a on day t, $E = 1000$ episodes.

2 Main result: with probability matching, either controller tips the swarm

Without a controller the swarm stays undecided. Under probability matching no episode ever becomes unanimous, and the A2 − A0 vote margin stays near zero (Figure 4, red): exactly 0 with full memory, and a small lean of +0.07 to +0.15 toward A2 when agents forget (A2-side facts survive forgetting better: held by 53% of agents on day 30, A0-side facts by 31%).

With a controller it tips within days. Figures 1 and 2 show the day-30 gain for every reading budget q_c and posting budget b; Figure 3 shows how the gain builds up over the days.

Table 1: Probability matching, day 30, $q_c = 12$, $b = 3$: paired gain in the target's share, Eq. (3), for $q = 3/6/12$; and the share of episodes in which the target passes 75% by day 30 (mean over q), with the silent value in brackets.

ρ	controller	gain ($q = 3/6/12$)	target past 75%	median day
0.75	truth (A0)	0.33 / 0.42 / 0.43	0.99 (0.26)	3–4
0.75	false (A2)	0.21 / 0.25 / 0.23	0.98 (0.58)	3
1.0	truth (A0)	0.37 / 0.37 / 0.38	1.00 (0.36)	3
1.0	false (A2)	0.08 / 0.07 / 0.06	0.90 (0.47)	3

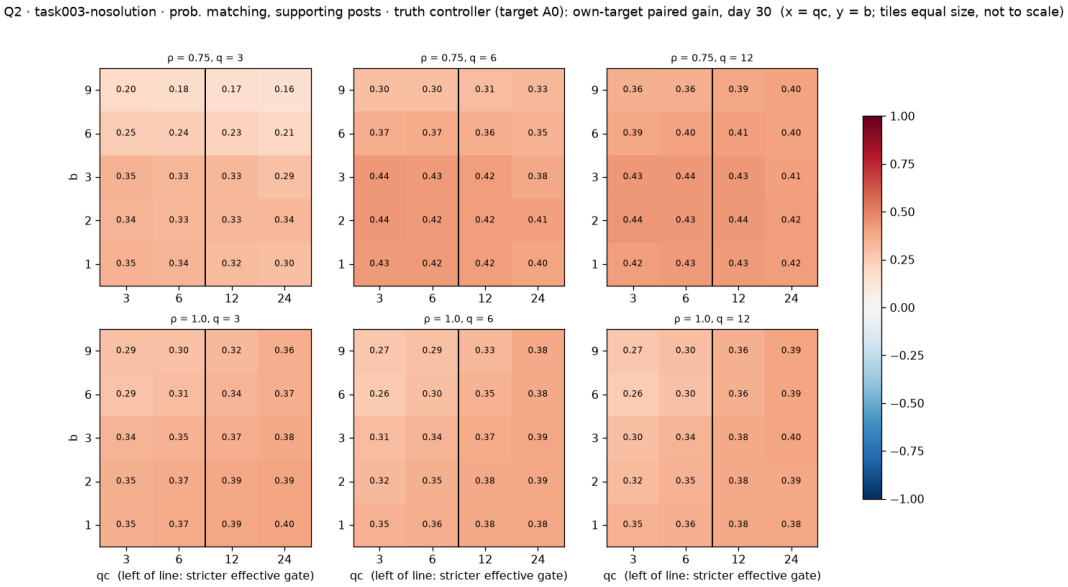


Figure 1: Truth controller, probability matching: day-30 paired gain in the A0 share over q_c (x) and b (y). Rows: ρ; columns: q. Tiles equal size, not to scale. Hatched: 95% interval includes 0. Left of the black line the vote gate is effectively stricter.

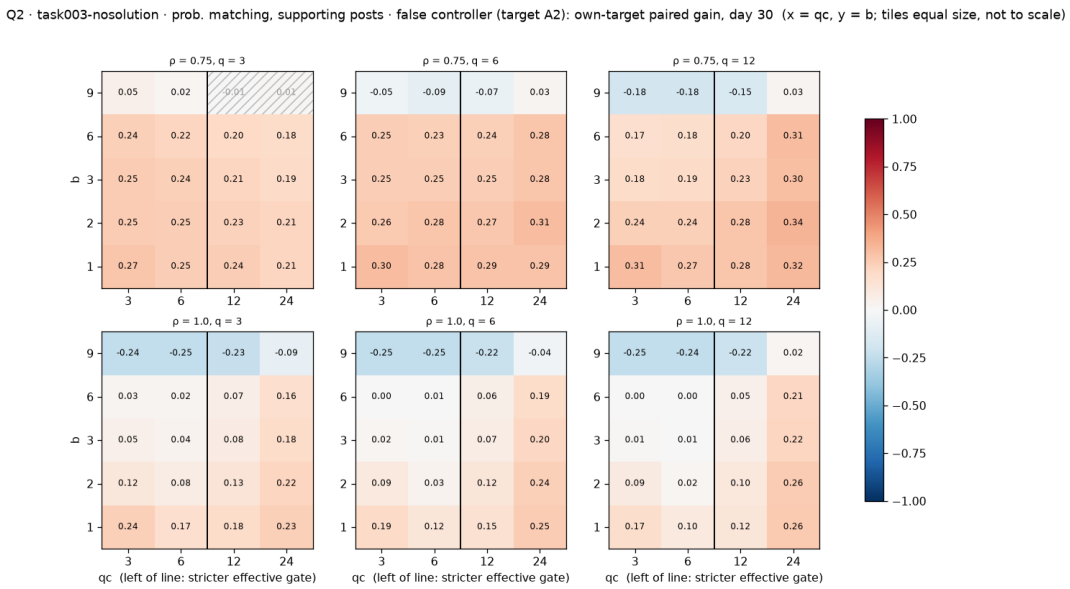


Figure 2: As Figure 1, for the false controller (gain in the A2 share).

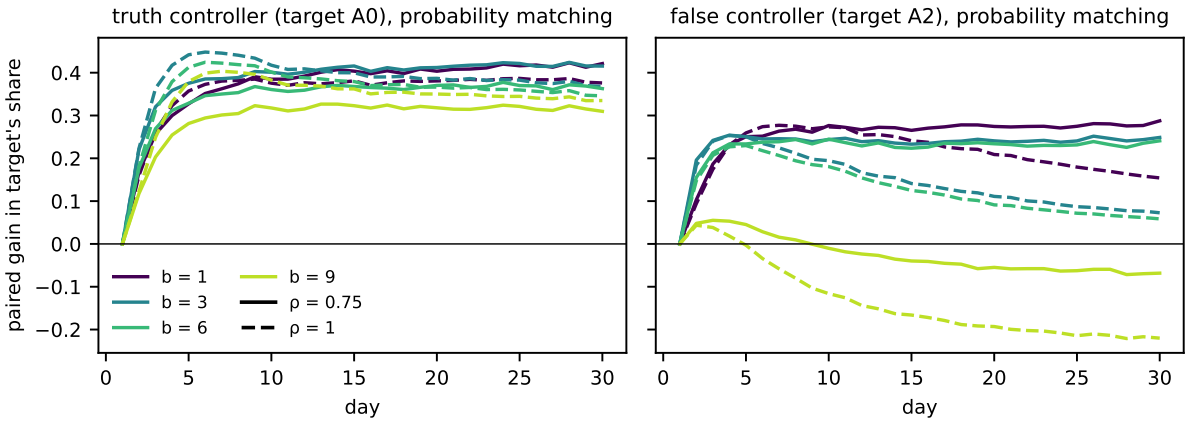


Figure 3: Probability matching: paired gain in the target's share over the days, by posting budget b; $q = 6$, $q_c = 12$. Solid $\rho = 0.75$, dashed $\rho = 1$.

- **Both controllers tip the swarm in about 3 days** (Table 1).
- **The truth controller is stronger**, and the gap widens with full memory: gains of about 0.37 against 0.07 at $\rho = 1$. As agents accumulate facts, their pooled evidence pulls back toward the truth's higher ceiling; the false controller's gain decays over the days (Figure 3, right, dashed).
- **The reading budget q_c matters little**; the posting budget matters only at its largest value (Section 4).

3 A special case: the A2 drift under argmax voting

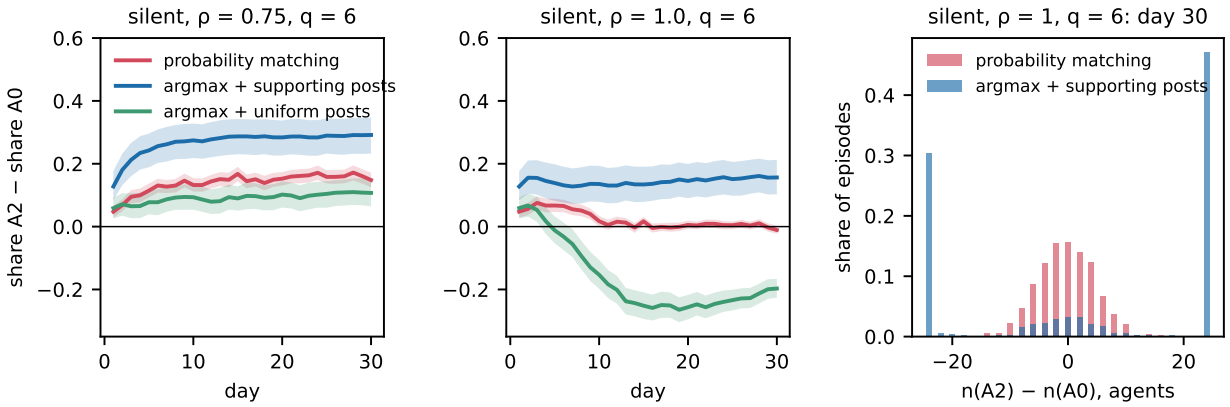


Figure 4: Silent arm, $q = 6$. Left and middle: A2 − A0 vote margin over the days. Right: day-30 spread of the margin across episodes, in agents.

With argmax voting and vote-supporting posts (the *base* rule), the silent swarm does not stay undecided: **63–91% of episodes end unanimous**, all 24 agents on A2 (43–67%) or on A0 (15–41%). The mean margin toward A2 (Figure 4, blue) is the difference between those two groups; the histogram shows the lock-in at ±24. Three ingredients combine:

1. **Argmax turns any lead into a vote**, Eq. (2).
2. **Agents post only facts supporting their vote**, so later readers mostly meet the majority's evidence, and with forgetting the minority's evidence dies out (day 30, $\rho = 0.75$: A2-side facts held by 53% of agents, A0-side by 24%).
3. **The swarm starts with more A2 voters.** Under argmax, an agent holding a fact that favours two allocations equally votes for each half the time. Its expected starting vote is

$$\mathbb{E}[n_k] = \sum_{j=1}^{24} \frac{\mathbb{1}\{k \in \arg\max P_j\}}{|\arg\max P_j|}, \tag{4}$$

   which gives **7 / 6 / 11** for A0 / A1 / A2 (Table 2).

Remove ingredient 1 (probability matching) or 2 (uniform posting: post any remembered fact) and the lock-in collapses: no unanimous episode at all under probability matching, 9–22% under uniform posting.

Table 2: Why the start is 7 / 6 / 11. The setup balanced agents by *fact group* (8 per allocation), assigning a tied fact to the lower-numbered allocation; argmax votes split ties.

	A0	A1	A2
agents per fact group (design bookkeeping)	8	8	8
of which hold a fact tied A1/A2 (counted as A1)	–	6	–
of which hold a fact tied A0/A1 (counted as A0)	2	–	–
expected starting votes, argmax, Eq. (4)	**7**	**6**	**11**
expected starting votes, probability matching	8.3	7.4	8.3

How to fix the start. Balance the agents on *expected votes*, Eq. (4), instead of on fact groups. Under the current rules this is impossible: none of the 10,438 coin-flip agent sets the builder can produce has equal expected A0 and A2 votes. The reason is scarcity: only 3 eligible facts favour A1 alone, so any 5-fact A1 group must include A1/A2-tied facts. Dropping the fact-group rule and requiring expected votes of exactly 8 / 8 / 8 directly works: a random search found such agent sets (still a coin flip, all facts eligible), e.g. 12 facts each held by 2 agents. A full rebuild would also re-check the pool rules (no shared facts, no joint proof). This changes the frozen setup, so it is a decision, not yet applied. Probability matching needs no fix: its start is already balanced.

4 More budget can be worse: a design effect

At the largest posting budget the controller's gain falls (Figures 1, 2, top row; Figure 3, $b = 9$), and the false controller's can even turn negative (-0.08 at $\rho = 0.75$, -0.22 at $\rho = 1$, $q_c = 12$, mean over q). The cause is arithmetic, not swarm dynamics. The controller must post exactly b facts or none, and the shared pool holds only n_{target} facts on its target's side, so each post contains at least

$$c = \max(0,\ b - n_{\text{target}}) \tag{5}$$

facts that do not favour its target.

Table 3: Pool composition and what the false controller actually posts (probability matching, $\rho = 1$, $q = 6$, $q_c = 12$). Tied facts count for both sides.

	$b = 3$	$b = 6$	$b = 9$
facts in the pool on the target's side (n_{target})		A0: 4, A2: 6	
posted facts favouring A0 / A1 / A2	14 / 23 / 62%	12 / 22 / 67%	**26** / 23 / 51%
nights the controller posts	59%	63%	**93%**

The fall starts exactly where Eq. (5) becomes positive: at $b = 6$ for the truth controller ($n = 4$), at $b = 9$ for the false one ($n = 6$). Its counter-evidence keeps the target's vote share below the gate, so it keeps posting (Table 3), and the counter-evidence accumulates. The effect replicates with fresh random draws (a second seed). It disappears if the controller may post *up to* b facts, or never posts facts against its target; in task003-symmetric, whose pools are entirely on their target's side, more budget simply helps.

5 The controller senses the swarm accurately

Each night the controller estimates the share of agents backing its target from the $n \le q_c$ agent posts it reads, $\hat{v} = \frac{1}{n}\sum_i \mathbb{1}\{\text{author of post } i \text{ voted for the target}\}$, and stays silent when $\hat{v} \ge 0.75$.

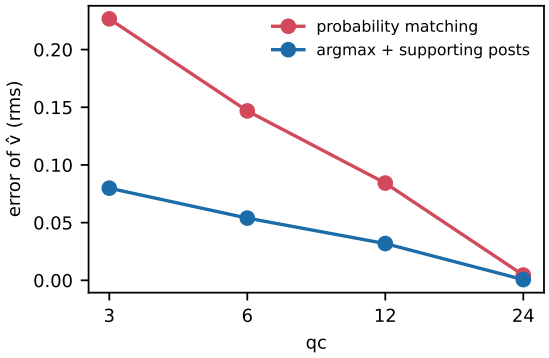


Figure 5: Error of $\hat{v}$ against the true share of all 24 agents, task003-nosolution.

$\hat{v}$ is unbiased (± 0.0005), its error shrinks with q_c and vanishes at $q_c = 24$ (Figure 5). It is larger under probability matching because those swarms stay mixed, which also makes gate mistakes more common: the controller stays silent although the true share is below 0.75 on 5–8% of nights, and posts although it is above 0.75 on up to 13% at $q_c = 3$. Abstaining agents (0.1–0.2% of agent-days) do not bias it.

Helping the truth versus agents' own proofs: does not apply here. In task003-symmetric, extra truth posts raise the share of agents holding a proof of A0 but lower the share holding one from their own evidence. In task003-nosolution no proof is possible by construction, so this question does not arise.

6 How efficiently does the controller steer?

A controller can change the swarm in two ways: in the direction it is asked to, or in ways that happen whatever its target. We measure how readable the command is in the swarm, how far each controller gets toward its target, how much of its change is directed, what each fact it posts buys, and how unsure the agents become. Throughout, the target is chosen by a fair coin, and all quantities are taken at one time t, across the 1,000 episodes of each arm (not over whole trajectories). Setting: probability matching, $q = 6$.

6.1 Measures

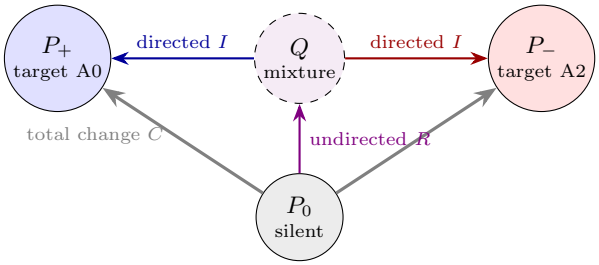


Diagram 2. Three distributions of the swarm's state at time t, one per arm, each over the episodes of that arm. The controller's total change C splits exactly into a part that depends on its target (I, from the mixture Q to each arm) and a part that does not (R, from silence to Q). Efficiency $\eta = I/C$ is the directed share.

Distributions. For each arm a (silent 0, target A0 +, target A2 −) and a time t, each episode e contributes a probability vector $y_a(e)$ over the values x of the state X: at *one-agent* level its vote shares (the vote of a randomly chosen agent), at *swarm* level a mark on the bin of $s = m_{\mathrm{A0}} - m_{\mathrm{A2}}$ (difference of mean beliefs, three equal bins on $[-1, 1]$). Then $\hat{P}_a = \frac{1}{n}\sum_e y_a(e)$. Any pair of controllers can be written as

$$\hat{P}_\pm = \hat{P}_0 + c \pm \Delta, \qquad c = \tfrac{1}{2}(\hat{P}_+ + \hat{P}_-) - \hat{P}_0, \quad \Delta = \tfrac{1}{2}(\hat{P}_+ - \hat{P}_-), \tag{6}$$

a *common* part c (what control does on average, whatever the target) and a *directional* part Δ.
Readability and the pair efficiency. With $\hat{Q} = \hat{P}_0 + c$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{Q}) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{Q}), \quad C = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{P}_0) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{P}_0) = I + R, \quad R = \mathrm{KL}(\hat{Q}\|\hat{P}_0), \quad \eta = I/C. \tag{7}$$

I is the mutual information between the target and the state: how readable the command is (at most 1 bit). I depends only on Δ, R only on c; the reference $\hat{P}_0$ is the silent arm. For small changes, $\mathrm{KL}(p + \epsilon\|p) \approx \sum_x \epsilon(x)^2/(2\ln 2\, p(x))$ gives

$$\eta \approx \frac{1}{1 + \|c\|^2/\|\Delta\|^2}, \qquad \|v\|^2 = \sum_x \frac{v(x)^2}{\hat{P}_0(x)}. \tag{8}$$

So $\eta = 1$ exactly when the two controllers' changes are mirror images ($c = 0$), whatever their size, and η falls only through the *ratio* of common to directional change: when both push the same way (a common push), or one moves the swarm much more than the other (a lopsided pair: along one direction with sizes a and b, $\|c\|/\|\Delta\| = |a-b|/(a+b)$). It is a property of the pair, not of either controller, and it stays constant whenever both effects grow together. At swarm level C is often infinite (the controlled swarm reaches states silence never visits); there we use $\eta_{JS} = I/(I + \mathrm{JS}(\hat{Q}, \hat{P}_0))$ with the bounded Jensen–Shannon divergence.
Each controller against its own target z (one agent, votes):

$$S_z = \mathrm{KL}\big(\mathrm{Bern}(\hat{P}_z(z)) \,\|\, \mathrm{Bern}(\hat{P}_0(z))\big), \qquad \pi_z = \frac{S_z}{\log_2[1/\hat{P}_0(z)]}, \qquad \varphi_z = \frac{S_z}{\mathrm{KL}(\hat{P}_z\|\hat{P}_0)}, \qquad \text{price} = \frac{S_z}{\text{facts posted}}. \tag{9}$$

S_z is the least information any intervention must inject to raise the target's share from its silent value to the observed one (raise the target, scale the other answers in proportion). **Progress** π_z is the share of the way from silence to unanimity on the target, in bits; **focus** φ_z the share of the controller's actual change that was needed for that gain; both are negative when the controller moved its target *down*. The **price** is the information injected per fact the controller posted. Finally, the agents' own uncertainty: $\Delta\bar{H}$ (mean entropy of their beliefs) and ΔJSD (disagreement, the entropy of the mean belief minus $\bar{H}$), controlled minus silent, same episode.

6.2 Over time, at one setting

- **Control is reached fast**: 90% of the day-30 progress by day 9 (target A0) and day 4 (target A2); $\pi_{\mathrm{A0}} = 0.37$, $\pi_{\mathrm{A2}} = 0.21$ at day 30, with focus 0.93 and 0.99.
- **Both controllers make agents surer** ($\Delta\bar{H} \approx -0.39$ bits); the truth controller also splits them a little (ΔJSD +0.06).

6.3 Across the control plane

- **The large-budget backfire is negative progress.** At $b = 9$ the false controller moves A2 *down* (π_{A2} down to -0.03 at $\rho = 0.75$ and -0.18 at $\rho = 1$) and its focus turns negative (-0.19 to -0.45 at $q_c \le 12$; -1 with full memory): its change goes against its own target. The truth controller keeps progressing (0.19–0.51).

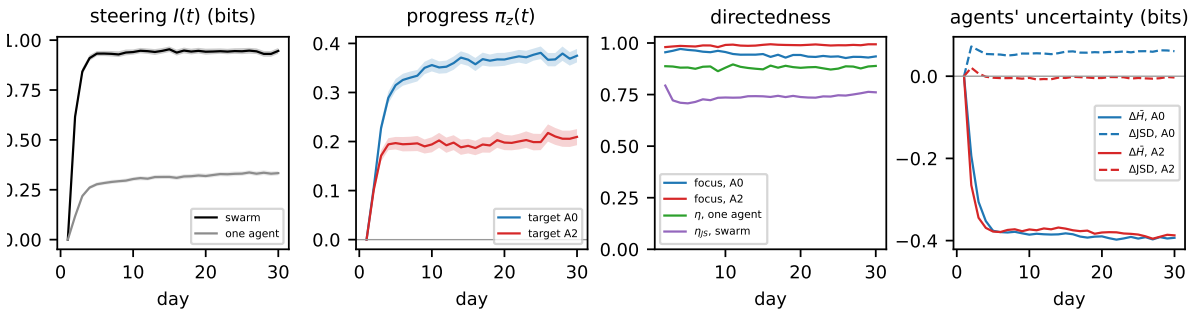


Figure 6: At $\rho = 0.75$, $q_c = 12$, $b = 3$ (end of each day): readability $I(t)$, each controller's progress $\pi_z(t)$, directedness (focus φ_z, pair efficiency η, η_{JS}), and the agents' uncertainty. Bands: 95% intervals over episodes.

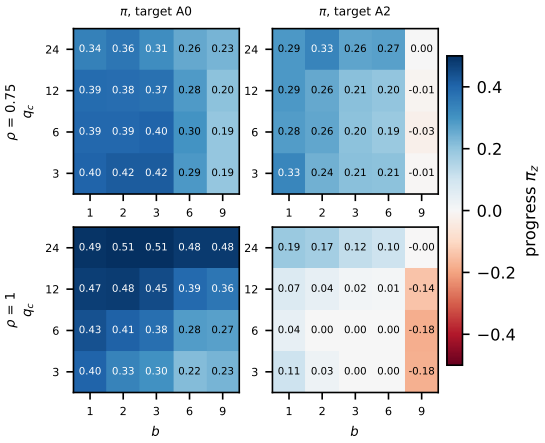


Figure 7: Progress π_z of each controller at $t = 30$ over the reading budget q_c and the facts per night b (one agent, votes). Red: the controller moved its target down. Dash: undefined (silence already at unanimity).

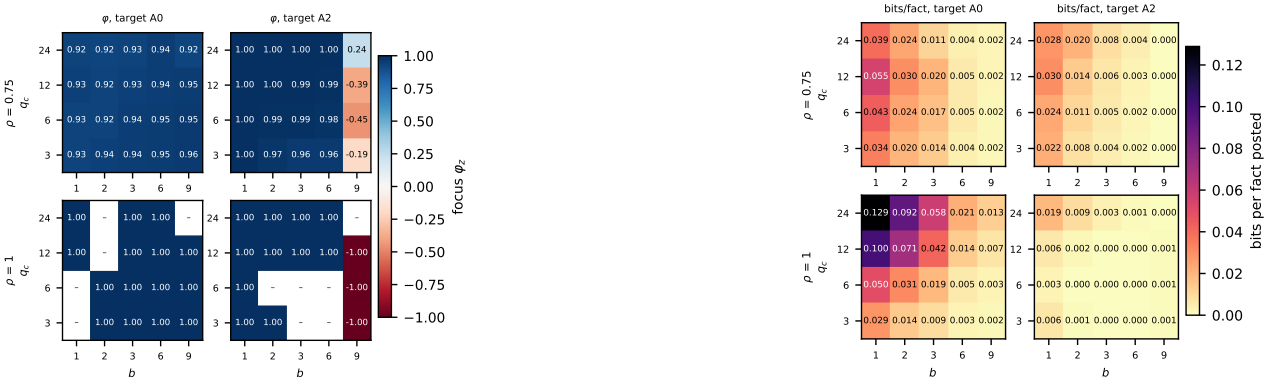


Figure 8: Left: focus φ_z (negative: the change went against the target). Right: price, information injected per fact posted (bits). $t = 30$; Dash: undefined (no change, or the controlled swarm considers an answer silence rules out completely).

- **The cheapest control is one fact per night**: the price (target A0) falls from 0.03–0.13 bits per fact at $b = 1$ to 0.002–0.013 at $b = 9$; with full memory, reading more posts makes each fact worth more (0.03 at $q_c = 3$, 0.13 at $q_c = 24$, $b = 1$).

6.4 Readability, pair efficiency and uncertainty against the facts per night

- η **collapses at** $b = 9$ (0.25 at $\rho = 0.75$, 0.06 at $\rho = 1$) through a *common push*: both controllers raise A0. Up to $b = 6$, at $\rho = 0.75$, the pair is balanced ($\eta \approx 0.9$).
- The swarm-level readability saturates near 1 bit with full memory (every episode ends on its controller's side), so it measures distinguishability, not size; progress measures size.

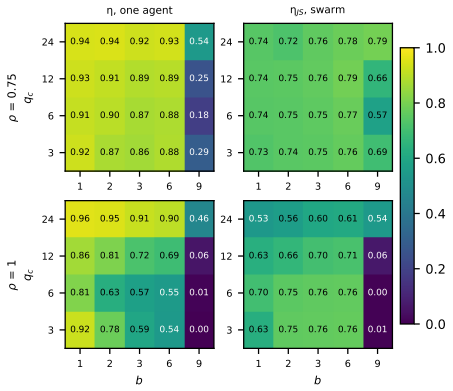


Figure 9: Pair efficiency at $t = 30$ over q_c and the facts per night b: one-agent η (beliefs) and swarm η_{JS}. Dash: undefined.

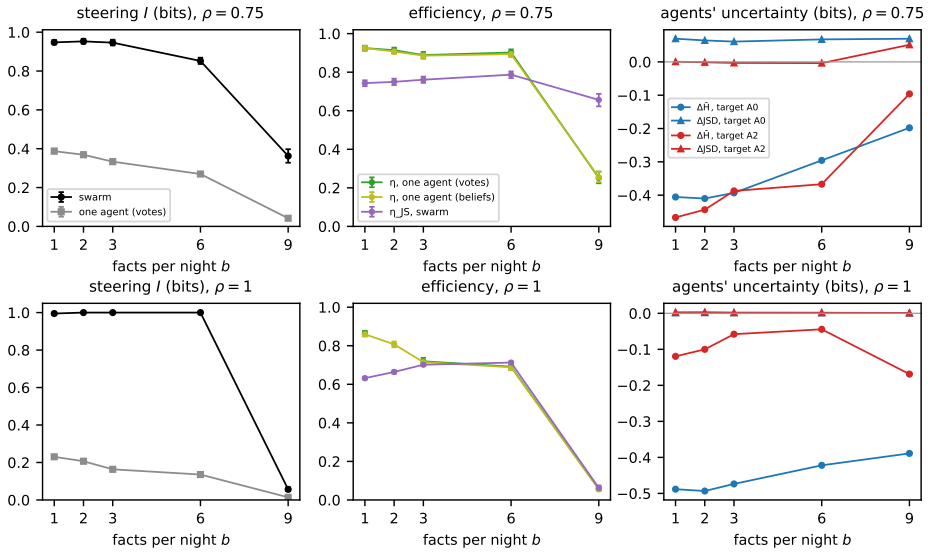


Figure 10: Readability I (left), pair efficiency (middle) and the change in the agents' uncertainty (right; circles $\Delta\bar{H}$, triangles ΔJSD) at $t = 30$ against facts per night b ($q_c = 12$). Bars: 95% intervals.

6.5 Precision at 50, 100 and 300 paths

Table 4: Value from 1,000 paths per arm, and root-mean-square error of the estimate from 50, 100 or 300 paths (200 random subsets each), at $t = 30$, $b = 3$, $q_c = 12$, $\rho = 0.75$. *: share of subsets in which the quantity is infinite or undefined.

quantity	1,000 paths	error, 50	error, 100	error, 300
steering I, swarm (3 bins)	0.946	0.034	0.024	0.012
steering I, one agent (votes)	0.333	0.027	0.019	0.009
efficiency η, one agent (votes)	0.889	0.038	0.027	0.013
efficiency η_{JS}, swarm	0.761	0.038	0.025	0.013
progress π, target A0 (votes)	0.375	0.034	0.020	0.010
progress π, target A2 (votes)	0.209	0.037	0.027	0.013
focus φ, target A0 (votes)	0.935	0.025	0.019	0.009
focus φ, target A2 (votes)	0.994	0.014	0.008	0.004
$\Delta\bar{H}$, target A0	-0.393	0.036	0.026	0.012
$\Delta\bar{H}$, target A2	-0.387	0.034	0.025	0.012
ΔJSD, target A0	0.061	0.010	0.006	0.004
ΔJSD, target A2	-0.003	0.009	0.006	0.003

Across all settings, at 100 paths: readability ± 0.01 (one agent) to ± 0.03 (swarm), progress ± 0.02, focus ± 0.01–0.02, η ± 0.02–0.03, $\Delta\bar{H}$ ± 0.02, ΔJSD ± 0.006. Two settings that differ by at least 0.02 are ordered correctly in 90–100% of subsets.

Caveats. One world (task_003), 24 agents, exact-Bayesian agents; means over 1,000 episodes per cell. Full analysis, code and tables: `analysis/task003_llm_free/sim1_analysis/` (reports Q0–Q6).